

The empirical recurrence for OEIS A251015: recovering a conjecture that is not written down

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Abstract

OEIS A251015 records “Empirical recurrence of order 92”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 243$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 92, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 92 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A251015 is “Number of $(n+1) \times (4+1)$ 0..2 arrays with no 2×2 subblock having the sum of its diagonal elements less than the maximum of its antidiagonal elements.”. It has offset 1 and begins

18910, 1691169, 151364054, 13508173078, 1205414504630, 107601568615993, 9606281958966442, 85763083692491

The entry states:

Empirical recurrence of order 92 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 12:59:28, revision 8) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 243$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 243$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 486$ exact terms of the model returns order 92 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{92} c_i a(n-i),$$

c_1	243	c_{24}	5046907553636221372971404	c_{47}	−22332415916984099215437
c_2	−25346	c_{25}	−132685627114742080044784944	c_{48}	21493905636102540993788
c_3	1609064	c_{26}	761487228667321577870232901	c_{49}	−18744031447432556961290
c_4	−71434779	c_{27}	−3178826613334523339388725860	c_{50}	14556702603384095005187
c_5	2384786541	c_{28}	11089511903871779964665442336	c_{51}	−9850362793960078650630
c_6	−62586855185	c_{29}	−33865429391771690537727419672	c_{52}	55798191207256329003790
c_7	1331174066722	c_{30}	92448873451365845878989019897	c_{53}	−2380048101173429124029
c_8	−23472670431211	c_{31}	−228082929320332914144684147545	c_{54}	4265346568364411973256
c_9	349321538646294	c_{32}	511653874001948744782699953573	c_{55}	4783061915318466039756
c_{10}	−4451640178970274	c_{33}	−1047239852305498065529763945618	c_{56}	−702720332421825722082
c_{11}	49161971409102236	c_{34}	1959353486513814451075411296415	c_{57}	5986543360748615878197
c_{12}	−475150453842814580	c_{35}	−3354144632355693743578941689415	c_{58}	−4040496222813778560650
c_{13}	4051633528180191693	c_{36}	5256251135142863202907685911019	c_{59}	23422354876966451428670
c_{14}	−30678505891353346556	c_{37}	−7545607026594690183711456367418	c_{60}	−120790119047373853486
c_{15}	207292350602068794098	c_{38}	9940546989807641217526123159753	c_{61}	5631513033912047447603
c_{16}	−1254074367119352450812	c_{39}	−12071638119832494902135530617013	c_{62}	−237156004308372692624
c_{17}	6803019171432235681257	c_{40}	13646405670372918443973140491033	c_{63}	879157555993676921339
c_{18}	−33061926039680678981780	c_{41}	−14630618839956000926881607315231	c_{64}	−26535918020746440226
c_{19}	143305359186872818819123	c_{42}	15319244453262991275174797030847	c_{65}	484004622490368225578
c_{20}	−548099601598362007935862	c_{43}	−16201099288678601720611761826988	c_{66}	94981684373288951181
c_{21}	1806335044798689230601918	c_{44}	17646633406832173573734081270927	c_{67}	−14907940472202019283
c_{22}	−4835256011500258968436073	c_{45}	−19593625791999526919705948414923	c_{68}	90213738488782775172
c_{23}	8519561182839295880608358	c_{46}	21451878021729401017196099700488	c_{69}	−3985494333380366995

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 92, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $92 + 4$ terms removed returns order 92 as well, so $\deg q_{\min} = 92 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 11 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $92 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 92 that this sequence satisfies — and that family turns out to have exactly one member.

References

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